

# Full Stack Development and Deployment of a Web-Based Speech Recognition System

Week 8  
2 March – 8 March 2026

Soh Zi Xiang  
Nanyang Technological University  
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# Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
  - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
  - Improve robustness, usability, and AI-driven recording analysis



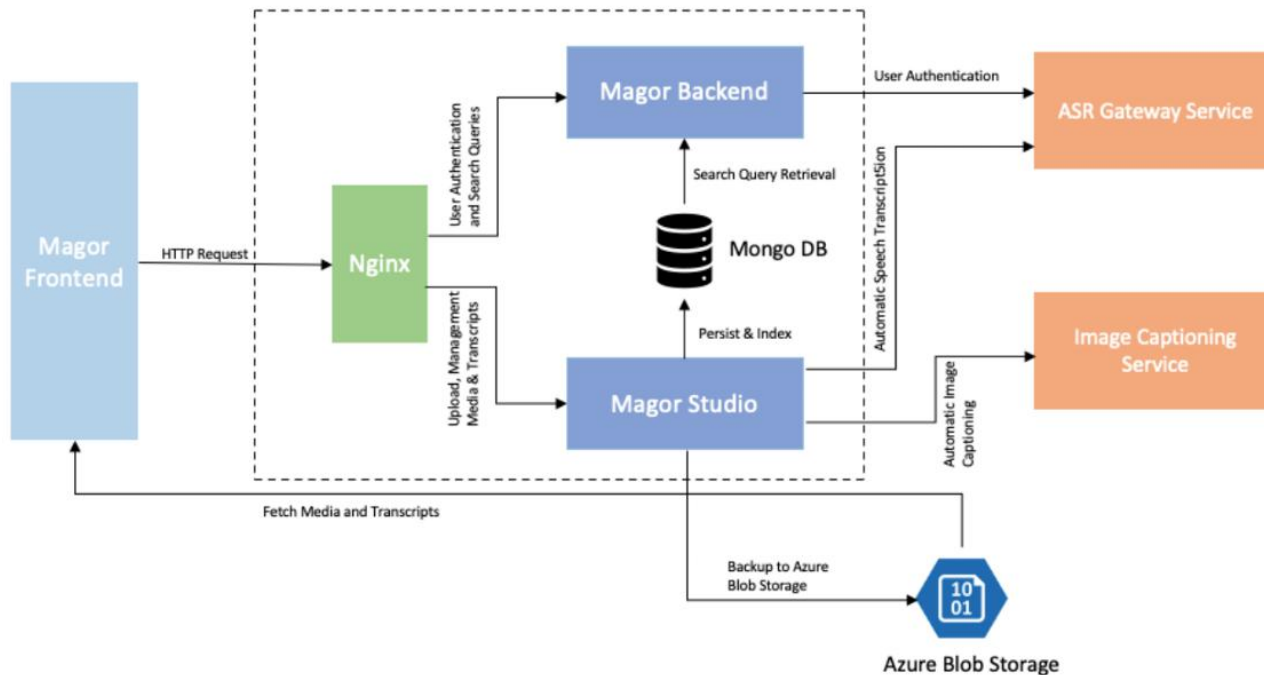
# Project Objectives

**The primary objectives of this project are:**

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



# Systems Overview



Source: Huang (2021)



# Agenda

## Work done this week:

1. Updated Transcript Studio UI
  - Adjust button order (Download after publish)
2. Removed empty tags from top transcript words
3. Restored admin page functionality (Recordings page)
  - Fixed upload date display and sorting
  - Fixed recording edit page
  - Fixed edit and update functionality



# Task 1 (Transcript Studio Button Reorder)

- Reordered button order for better workflow

The screenshot displays the Transcript Studio interface. At the top, a blue header bar contains the name 'Magor' and a dropdown menu set to 'test'. Below the header, the left sidebar shows the 'Title' field with the text 'Indonesian stocks plunges' and the 'Description' field with 'No Description'. A 'Transcript ID: 69895c...' is visible, along with a 'Go to Recording' button. The main area features a video player with a thumbnail for 'ASIA Business Tonight' and playback controls. Below the video player, a 'Transcripts' table is shown, sorted by 'Created Date (Earliest)'. The table lists two transcripts. The first transcript, ID '69818288c1ac0900232a9377', is 'Auto Generated' and has 'Edit', 'Download', and 'Delete' buttons. The second transcript, ID '69895ce4b0342000233b6c4c', is 'Uploaded' and has 'Edit', 'Download', 'Publish', and 'Delete' buttons. A red box highlights the 'Transcripts' table area. To the right of the table, a 'Delete' button is visible. A speaker icon is located in the bottom right corner of the interface.

Magor test

Title  
Indonesian stocks plunges

Description  
No Description

Transcript ID: 69895c... [Go to Recording](#)

Uploaded

[Edit](#) [Download](#)

**Transcripts** [Sort By Created Date \(Earliest\)](#)

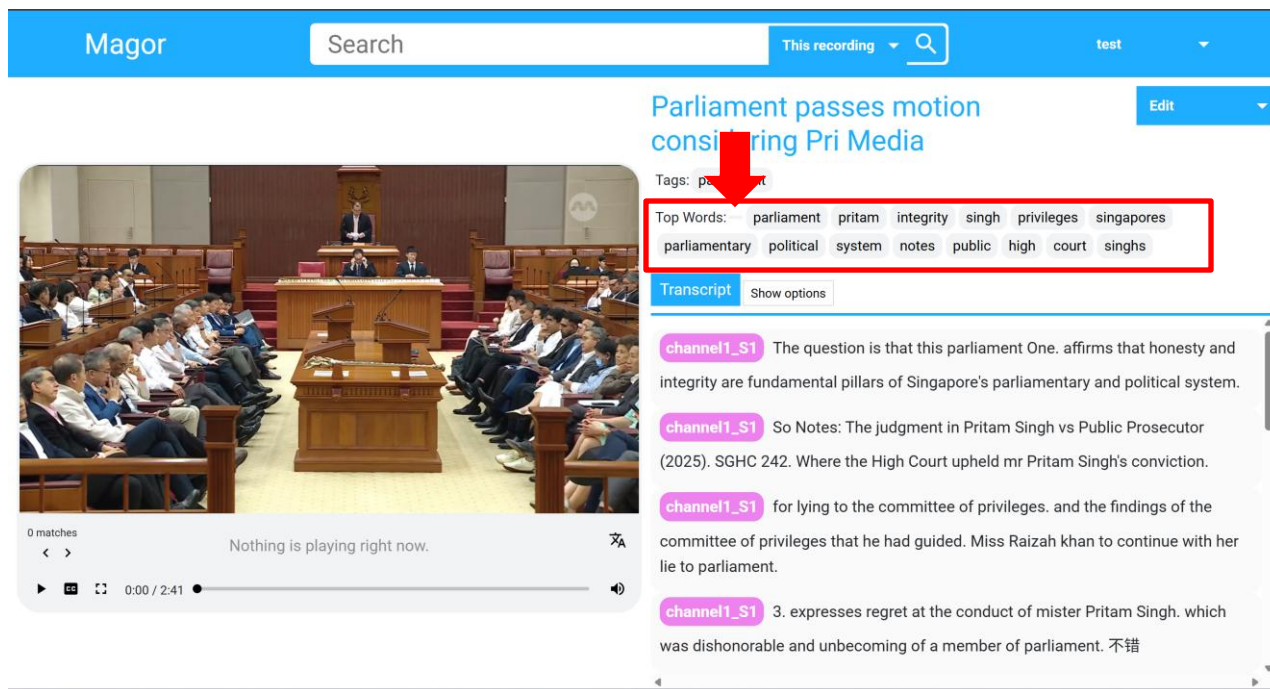
|                                                                                              |                         |                              |
|----------------------------------------------------------------------------------------------|-------------------------|------------------------------|
| Transcript ID: 69818288c1ac0900232a9377                                                      | Created At: a month ago | Last Updated At: a month ago |
| Auto Generated                                                                               |                         |                              |
| <a href="#">Edit</a> <a href="#">Download</a> <a href="#">Delete</a>                         |                         |                              |
| Transcript ID: 69895ce4b0342000233b6c4c                                                      | Created At: a month ago | Last Updated At: 6 days ago  |
| Uploaded                                                                                     |                         |                              |
| <a href="#">Edit</a> <a href="#">Download</a> <a href="#">Publish</a> <a href="#">Delete</a> |                         |                              |

[Delete](#)

Speaker icon

## Task 2 (Removed Empty Tags from Top Words)

- Removed empty tags for cleaner top transcript words



Magor Search This recording test Edit

### Parliament passes motion considering Pri Media

Tags: parliament pritam integrity singh privileges singapore parliamentary political system notes public high court singhs

Transcript Show options

channel1\_S1 The question is that this parliament One. affirms that honesty and integrity are fundamental pillars of Singapore's parliamentary and political system.

channel1\_S1 So Notes: The judgment in Pritam Singh vs Public Prosecutor (2025). SGHC 242. Where the High Court upheld mr Pritam Singh's conviction.

channel1\_S1 for lying to the committee of privileges. and the findings of the committee of privileges that he had guided. Miss Raizah khan to continue with her lie to parliament.

channel1\_S1 3. expresses regret at the conduct of mister Pritam Singh. which was dishonorable and unbecoming of a member of parliament. 不错

0 matches < > Nothing is playing right now. 0:00 / 2:41

## Task 2 (Removed Empty Tags from Top Words)

- Filter empty tags to select only relevant top words

```
<FlexWrap className="recordingTagsContainer">
  Top Words:
  {topWords
    .filter((t) => t?.[0]?.trim())
    .map((t) => {
      const topWord = t[0]
      return (
        <Tag
          onClick={() => performSearch(topWord)}
          key={_key++}
          name={topWord}
        />
      )
    })
  }
  Filter when tokenizing speech segments
</FlexWrap>
```



Frontend Filter





# Task 3 (Restore Admin Recording Page Functionality)

## 1. Fixed upload date display and sorting

Recordings

Users

recordings

asrRequests

Statistics

Click on any field header to sort by that field

Filter by title, description, tags:  [APPLY](#)

☐

Title

Description

Path

Uploader

Upload date

↓

☐

speech video test

test

[EDIT](#) [DELETE](#) [View](#)

☐

harvard audio

test

[EDIT](#) [DELETE](#) [View](#)

☐

Indonesian stocks plunges

test

[EDIT](#) [DELETE](#) [View](#)

☐

Parliament passes motion considering Pri Media

test

[EDIT](#) [DELETE](#) [View](#)

☐

Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4

test

[EDIT](#) [DELETE](#) [View](#)

[+ CREATE](#) [EXPORT](#)

Rows per page: 10 1-5 of 5



# Task 3 (Restore Admin Recording Page Functionality)

## 1. Fixed upload date display and sorting

- Fixed data mapping mismatch

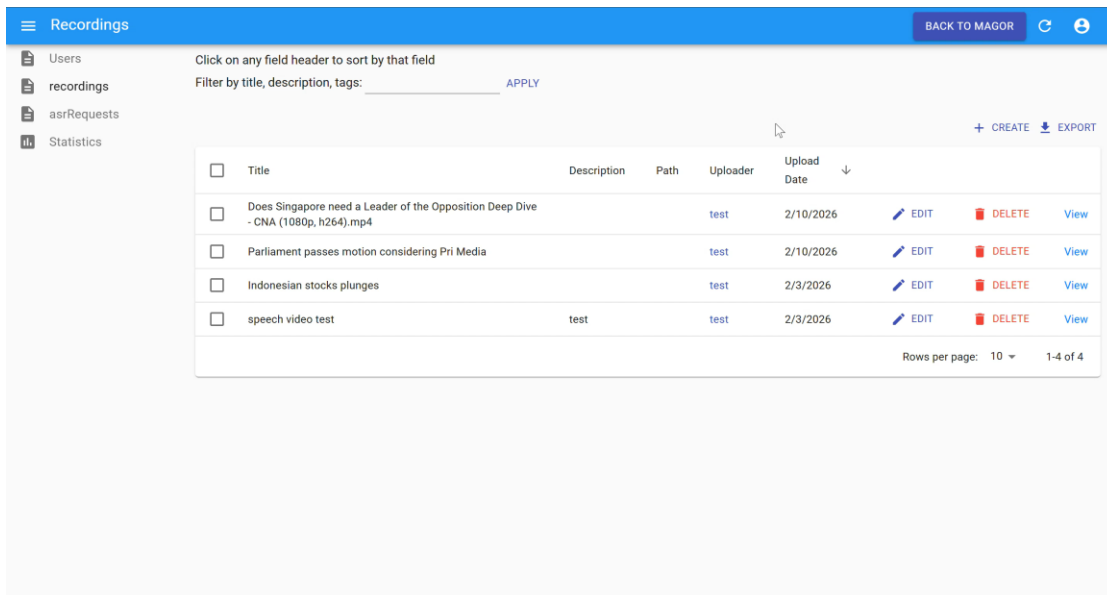
```
<List
  {...props}
  sort={{ field: 'uploadDate', order: 'DESC' }}
  filter={{
    key: filter,
    fields: ['title', 'description', 'tags.tagName'],
  }}
>
  <Datagrid rowClick="edit">
    <TextField source="title" />
    <TextField source="description" />
    <TextField source="path" />
    <ReferenceField source="uploaderId" reference="users">
      <TextField source="name" />
    </ReferenceField>
    <DateField source="uploadDate" />
    <EditButton />
    <DeleteButton />
    <RecordingLinkField />
  </Datagrid>
</List>
```



```
<List
  {...props}
  sort={{ field: 'createdAt', order: 'DESC' }}
  filter={{
    key: filter,
    fields: ['title', 'description', 'tags.tagName'],
  }}
>
  <Datagrid rowClick="edit">
    <TextField source="title" />
    <TextField source="description" />
    <TextField source="path" />
    <ReferenceField source="uploaderId" reference="users">
      <TextField source="name" />
    </ReferenceField>
    <DateField source="createdAt" />
    <EditButton />
    <DeleteButton />
    <RecordingLinkField />
  </Datagrid>
</List>
```

# Task 3 (Restore Admin Recording Page Functionality)

## 1. Fixed upload date display and sorting



The screenshot displays the 'Recordings' page in an admin interface. The page has a blue header with a 'BACK TO MAGOR' button and a sidebar with navigation links: Users, recordings, asrRequests, and Statistics. The main content area shows a table of recordings with columns for Title, Description, Path, Uploader, and Upload Date. The 'Upload Date' column is highlighted, indicating it is the current sort order. The table contains four rows of recordings, all with an upload date of 2/10/2026 or 2/3/2026. Each row has 'EDIT', 'DELETE', and 'View' action buttons. A filter bar at the top allows filtering by title, description, or tags. The bottom of the table shows 'Rows per page: 10' and '1-4 of 4'.

Recordings

Click on any field header to sort by that field

Filter by title, description, tags:  [APPLY](#)

+ CREATE [EXPORT](#)

| <input type="checkbox"/> | Title                                                                            | Description | Path | Uploader | Upload Date | ↓ |                                                                  |
|--------------------------|----------------------------------------------------------------------------------|-------------|------|----------|-------------|---|------------------------------------------------------------------|
| <input type="checkbox"/> | Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4 |             |      | test     | 2/10/2026   |   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | Parliament passes motion considering Pri Media                                   |             |      | test     | 2/10/2026   |   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | Indonesian stocks plunges                                                        |             |      | test     | 2/3/2026    |   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | speech video test                                                                | test        |      | test     | 2/3/2026    |   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |

Rows per page: 10 1-4 of 4



# Task 3 (Restore Admin Recording Page Functionality)

## 2. Fixed edit recordings data fetching

**Edit User** [BACK TO MAGOR](#)

- Users
- recordings**
- asrRequests
- Statistics

**Title**  
Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4

**Description**  
Podcast by CNA

**Recording Path**

**Transcripts**  
[ADD](#)

**Image Captions**  
[ADD](#)

**Sound Captions**  
Uploader: test

**Upload Date**  
mm/dd/yyyy

[SAVE](#) [DELETE](#)



## Task 3 (Restore Admin Recording Page Functionality)

### 2. Fixed edit recordings data fetching

```
<DateInput  
  source="uploadDate"  
  disabled={true}  
  label="Upload Date"  
>
```

```
<DateInput  
  source="createdAt"  
  disabled={true}  
  label="Upload Date"  
>
```

Date Mapping Correction  
Path Mapping Correction



# Task 3 (Restore Admin Recording Page Functionality)

## 2. Fixed edit recordings data fetching

Edit User

BACK TO MAGOR

Users

recordings

asrRequests

Statistics

Title

Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4

Description

Podcast by CNA

Recording Path

/usr/src/app/public/recordings/6d2c06d3-ceee-4235-9855-e68d10e9db2a.mp4

Transcripts

ADD

Image Captions

ADD

Sound Captions

Uploader

test

Upload Date

02/10/2026

SAVE

DELETE



# Task 3 (Restore Admin Recording Page Functionality)

## 3. Fixed edit and update functionality (Delete Issues)

Recordings

Users

recordings

asrRequests

Statistics

Click on any field header to sort by that field

Filter by title, description, tags:  [APPLY](#)

[+ CREATE](#) [↓ EXPORT](#)

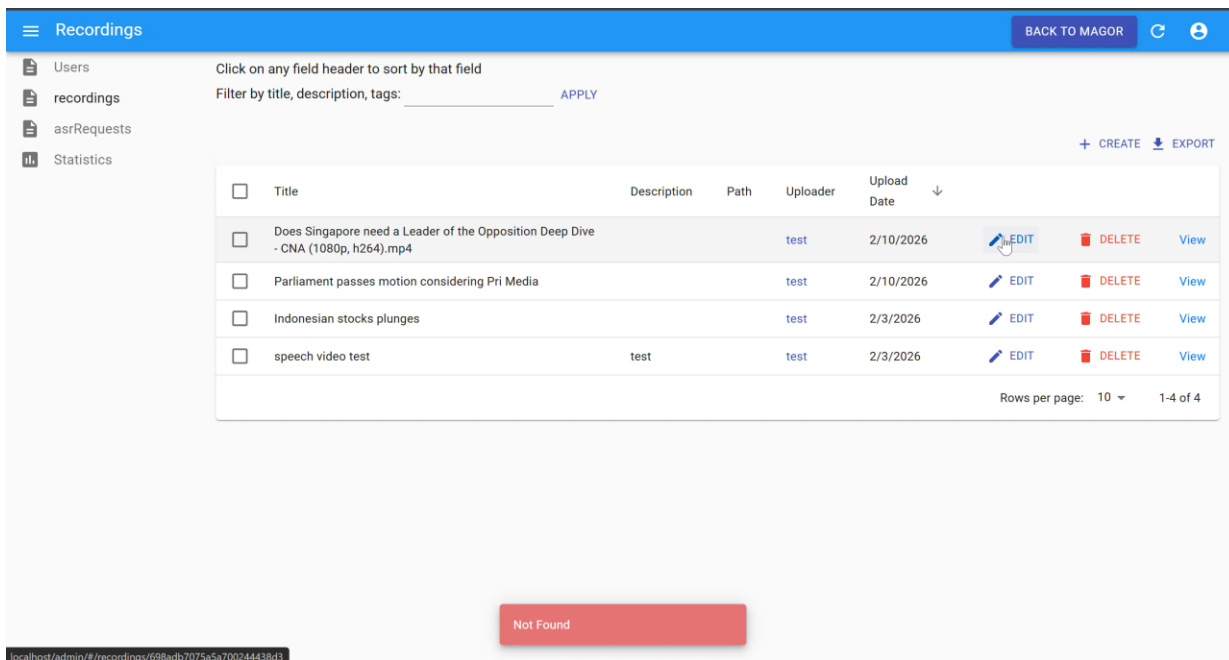
| <input type="checkbox"/> | Title                                                                            | Description | Path | Uploader | Created at | ↓                                                                |
|--------------------------|----------------------------------------------------------------------------------|-------------|------|----------|------------|------------------------------------------------------------------|
| <input type="checkbox"/> | Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4 |             |      | test     | 2/10/2026  | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | Parliament passes motion considering Pri Media                                   |             |      | test     | 2/10/2026  | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | Indonesian stocks plunges                                                        |             |      | test     | 2/3/2026   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | harvard audio                                                                    |             |      | test     | 2/3/2026   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | speech video test                                                                |             |      | test     | 2/3/2026   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |

Rows per page: 10 1-5 of 5



# Task 3 (Restore Admin Recording Page Functionality)

## 3. Fixed edit and update functionality (Description Update Issues)



The screenshot displays the 'Recordings' management page. On the left is a sidebar with navigation links: Users, recordings, asrRequests, and Statistics. The main area features a header with a filter input and an 'APPLY' button. Below this is a table of recordings. Each row includes a checkbox, a title, a description, a path, an uploader, an upload date, and three action buttons: EDIT, DELETE, and View. The table shows four recordings. At the bottom right of the table area, it indicates 'Rows per page: 10' and '1-4 of 4'. A red 'Not Found' message box is visible at the bottom center of the page.

| <input type="checkbox"/> | Title                                                                            | Description | Path | Uploader | Upload Date |                                                                  |
|--------------------------|----------------------------------------------------------------------------------|-------------|------|----------|-------------|------------------------------------------------------------------|
| <input type="checkbox"/> | Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4 |             |      | test     | 2/10/2026   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | Parliament passes motion considering Pri Media                                   |             |      | test     | 2/10/2026   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | Indonesian stocks plunges                                                        |             |      | test     | 2/3/2026    | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | speech video test                                                                | test        |      | test     | 2/3/2026    | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |

Rows per page: 10 1-4 of 4

Not Found





## Task 3 (Restore Admin Recording Page Functionality)

### 3. Fixed edit and update functionality (Description Update Issues)

```
698- // exports.updateStatus = async (req, res) => {
699- //   logger.info("Received status update:", req.body._id, req.body.stat
700- //   if (req.body.status != "done") {
701- //     res.send(true);
702- //     return;
703- //   }
704- //   try {
705- //     const asrId = req.body._id;
706- //     const link = req.body.result;
707- //     await fetchAndUpdateASR(asrId, link);
708- //   } catch (err) {
709- //     logger.error("ERROR IN UPDATE STATUS", err);
710- //   } finally {
711- //     res.send(true);
712- //   }
713- // };

713+ exports.updateStatus = async (req, res) => {
714+   logger.info("Received status update:", req.body._id, req.body.stat
715+   if (req.body.status != "done") {
716+     res.send(true);
717+     return;
718+   }
719+   try {
720+     const asrId = req.body._id;
721+     const link = req.body.result;
722+     await fetchAndUpdateASR(asrId, link);
723+   } catch (err) {
724+     logger.error("ERROR IN UPDATE STATUS", err);
725+   } finally {
726+     res.send(true);
727+   }
728+ };
```

Backend functionality was commented and  
some codes not aligned with current data  
formats



# Task 3 (Restore Admin Recording Page Functionality)

## 3. Fixed edit and update functionality (Delete Fixed)

Recordings

Users

recordings

asrRequests

Statistics

Click on any field header to sort by that field

Filter by title, description, tags:  [APPLY](#)

[+ CREATE](#) [↓ EXPORT](#)

| <input type="checkbox"/> | Title                                                                            | Description | Path | Uploader | Created at | ↓ |                                                                  |
|--------------------------|----------------------------------------------------------------------------------|-------------|------|----------|------------|---|------------------------------------------------------------------|
| <input type="checkbox"/> | Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4 |             |      | test     | 2/10/2026  |   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | Parliament passes motion considering Pri Media                                   |             |      | test     | 2/10/2026  |   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | Indonesian stocks plunges                                                        |             |      | test     | 2/3/2026   |   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | harvard audio                                                                    |             |      | test     | 2/3/2026   |   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | speech video test                                                                |             |      | test     | 2/3/2026   |   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |

Rows per page: 10 1-5 of 5



# Task 3 (Restore Admin Recording Page Functionality)

## 3. Fixed edit and update functionality (Description Update Fixed)

Recordings

BACK TO MAGOR

Users

recordings

asrRequests

Statistics

Click on any field header to sort by that field  
Filter by title, description, tags:  [APPLY](#)

+ CREATE

EXPORT

| <input type="checkbox"/> | Title                                                                            | Description | Path | Uploader | Upload Date | ↓ |                                                                  |
|--------------------------|----------------------------------------------------------------------------------|-------------|------|----------|-------------|---|------------------------------------------------------------------|
| <input type="checkbox"/> | Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4 |             |      | test     | 2/10/2026   |   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | Parliament passes motion considering Pri Media                                   |             |      | test     | 2/10/2026   |   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | Indonesian stocks plunges                                                        |             |      | test     | 2/3/2026    |   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |
| <input type="checkbox"/> | speech video test                                                                | test        |      | test     | 2/3/2026    |   | <a href="#">EDIT</a> <a href="#">DELETE</a> <a href="#">View</a> |

Rows per page: 10 1-4 of 4



# Plans for Next Week

- Restore admin page functionality
  - Recording Management Page
    - Fix path display and table overflows
  - User Management Page
    - Fix user creation errors
    - Resolve status update errors for unverified users
    - Improve user feedback messages (errors and success notifications)
    - Implement email and password format validation on frontend

